

Learning Quantum Processes from Quantum Statistical Queries: A Reproduction and Extension Study

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Video presentation: [\[YouTube unlisted link --- replace before submission\]](#)

Abstract

We study Wadhwa and Doosti’s recent paper *Learning Quantum Processes with Quantum Statistical Queries* [4]. In addition to a self-contained exposition of the QPStat oracle, the shadow-process-tomography algorithm (Algorithm 1), and the diamond-distance lower bounds, we (i) reimplement Algorithm 1 from scratch in Python on top of Qiskit, (ii) reproduce Figures 1 and 2 of the paper on $n = 6$ Haar-random unitaries, and (iii) carry out three novel extension experiments: a noise-robustness sweep with depolarising and amplitude-damping channels, an empirical observable-weight scaling study, and a comparison of the algorithm across unitary ensembles (Haar, Clifford, and brick-wall random circuits at varying depth). Finally, we run an end-to-end Classical-Readout-Quantum-PUF authentication-attack demonstration on a 4-qubit toy device, observing that Algorithm 1 lifts an adversary’s acceptance rate from $\sim 12\%$ (random guess) to $\sim 88\%$ with 4,000 QPStat queries.

1 Introduction

Motivation. Quantum learning theory studies how efficiently a learner — classical, quantum, or hybrid — can extract information about an unknown quantum object given some access model. The bulk of the literature has focused on *quantum states* as both the input and the target of learning. Wadhwa and Doosti instead consider *quantum processes* (CPTP maps) in a single-copy, no-entangled-ancilla access model that closely resembles what near-term hardware actually exposes.

Their access model is a natural quantum generalisation of Kearns’ classical *statistical query* (SQ) framework [3]: the learner submits an input state ρ together with an observable O and a tolerance τ , and receives an estimate $\alpha \approx \text{tr}(O\mathcal{E}(\rho))$ such that $|\alpha - \text{tr}(O\mathcal{E}(\rho))| \leq \tau$ with high probability.

Contributions of the paper. The paper proves four main results:

1. An efficient (quasipolynomial-query) algorithm for *average-case shadow process tomography* from QPSQs (Theorem 1).
2. A nearly matching $\Omega(M)$ lower bound in the number of observables.
3. An exponential lower bound for learning unitary 2-designs and a *doubly* exponential lower bound for Haar-random unitaries with respect to diamond distance (Theorems 2–4), via a many-vs-one distinguishing reduction.
4. A learning-based attack against authentication protocols built on Classical-Readout Quantum Physically Unclonable Functions (CR-QPUFs).

Our contributions. This report goes beyond a paper summary in three ways:

- **Reimplementation.** We reimplement the QPStat oracle and Algorithm 1 from scratch in a uv-managed Python project, with unit tests for endian conventions, oracle calibration, and ground-truth recovery on identity and Pauli- X processes (Section 4).
- **Reproductions and extensions.** We reproduce Figures 1 and 2 of the paper, then run three experiments that the paper does not (noisy-channel robustness, observable weight scaling, and circuit-class comparison; Section 5).
- **End-to-end CR-QPUF demonstration and defense.** We instantiate a 6-qubit CR-QPUF, build the authentication protocol, and run Algorithm 1 as the adversary, measuring acceptance rate vs. adversary query budget. We then instantiate the biased-response defense left open by the paper’s Remark 4 — a secret weight- w sign-parity bias — and show it pins a degree- $k < w$ QPSQ adversary at the random-guess floor while leaving honest authentication intact (Section 6).

Roadmap. Section 2 fixes notation and recalls the necessary background. Section 3 formalises the QPStat oracle and restates Algorithm 1. Section 4 covers the reproduction of Figures 1–2. Section 5 presents the three extension experiments. Section 6 gives the CR-QPUF attack and our sign-parity defense. Section 7 sketches the diamond-distance lower bounds. Section 8 discusses limitations and open questions.

2 Background

Pauli operators and stabilizer product states. We work with n -qubit systems of dimension $N = 2^n$. The single-qubit Pauli set is $\mathcal{P}_1 = \{I, X, Y, Z\}$ and the n -qubit *Pauli strings* $\mathcal{P}_n = \mathcal{P}_1^{\otimes n}$ form an orthonormal basis of $\mathcal{M}_{N,N}$ under the Hilbert-Schmidt inner product. The *degree* $|P|$ of a Pauli string is the number of non-identity factors. A *stabilizer product state* $|\psi\rangle = \bigotimes_{i=1}^n |s_i\rangle$ has each $|s_i\rangle \in \text{stab}_1 = \{|0\rangle, |1\rangle, |+\rangle, |-\rangle, | + y\rangle, | - y\rangle\}$. There are 6^n such states, and the uniform distribution over them is the *stabilizer-product-state design* used in Algorithm 1.

Haar measure and t -designs. Let μ_H be the unique left-and-right invariant probability measure on $\mathcal{U}(N)$. A distribution ν over $\mathcal{U}(N)$ is a *unitary t -design* iff its t -th moment super-operator equals that of μ_H . The Clifford group is a 3-design [5], and brick-wall random circuits with sufficient depth approximate t -designs [1].

Diamond distance. For two CPTP maps $\mathcal{E}_1, \mathcal{E}_2$, $d_{\diamond(\mathcal{E}_1, \mathcal{E}_2)} = \frac{1}{2} \|\mathcal{E}_1 - \mathcal{E}_2\|_{\diamond}$ where $\|\mathcal{E}\|_{\diamond} = \max_{\rho} \|(\mathcal{E} \otimes \mathcal{I})(\rho)\|_1$.

Classical shadows. Huang, Kueng and Preskill’s classical shadow framework [2] estimates many properties of an unknown state from a small number of randomized Pauli measurements. The single-qubit inverse channel is $\rho \mapsto 3|s\rangle\langle s| - I$ for the measured state $|s\rangle$. We use this only as a baseline against the calibrated Gaussian-noise emulator of QPStat in Figure 1.

3 The QPStat Oracle and Algorithm 1

3.1 Definition of the oracle

Definition 1 (QPStat oracle, [4]). *Given a quantum process $\mathcal{E} : \mathcal{M}_{N,N} \rightarrow \mathcal{M}_{N,N}$, the QPStat oracle takes as input an observable O with $\|O\|_{\infty} \leq 1$, a state $\rho \in \mathcal{S}_N$, and a tolerance $\tau \geq 0$,*

and returns a number α satisfying

$$|\alpha - \text{tr}(O \mathcal{E}(\rho))| \leq \tau \quad \text{with probability at least } 1 - \delta. \quad (1)$$

In our reimplementaion we provide two emulators of QPStat: a *calibrated Gaussian* oracle that adds normal noise of standard deviation $\sigma = \tau/\Phi^{-1}(1 - \delta/2)$ to the analytically computed expectation value (matching the paper’s simulation methodology), and a *shot-based* oracle that samples each Pauli basis the requisite number of times. The Gaussian emulator is a worst-case proxy [2]: any practical near-term device should achieve smaller deviations.

3.2 Algorithm 1

The algorithm proceeds in three stages, with hyperparameters

$$k = \lceil \log_{1.5}(2/\epsilon^2) \rceil, \quad \tilde{\epsilon} = \Theta\left(\frac{\epsilon^2}{(2n)^k}\right). \quad (2)$$

Algorithm 1 Average-case shadow process tomography from QPSQs (paper Algorithm 1)

```

1: for  $\ell = 1, \dots, N$  do
2:   Sample  $|\psi_\ell^{(\text{in})}\rangle$  uniformly from  $\text{stab}_1^{\otimes n}$ .
3:    $y_\ell \leftarrow \text{QPStat}_{\mathcal{E}}(|\psi_\ell^{(\text{in})}\rangle\langle\psi_\ell^{(\text{in})}|, O, \tau)$ .
4: end for
5: for each Pauli  $P \in \mathcal{P}_n$  with  $|P| \leq k$  do
6:    $\hat{x}_P(O) \leftarrow \frac{1}{N} \sum_{\ell=1}^N y_\ell \text{tr}(P |\psi_\ell^{(\text{in})}\rangle\langle\psi_\ell^{(\text{in})}|)$ .
7:   if  $(1/3)^{|P|} > 2\tilde{\epsilon}$  and  $|\hat{x}_P(O)| > 2 \cdot 3^{|P|/2} \sqrt{\tilde{\epsilon}} \|O\|_{\text{Pauli},1}$  then
8:      $\hat{\alpha}_P(O) \leftarrow 3^{|P|} \hat{x}_P(O)$ .
9:   else
10:     $\hat{\alpha}_P(O) \leftarrow 0$ .
11:   end if
12: end for
13: return  $h(\rho) = \sum_{P: |P| \leq k} \hat{\alpha}_P(O) \text{tr}(P\rho)$ .
```

Implementation note: 64× speedup. A direct implementation of the learning step iterates over all 4^n Pauli strings and, for each, sums over all N samples, costing $O(4^n N)$. We observe that for any stabilizer product state, only 2^n Paulis (those whose non-identity factors match the stabilizer axes on each qubit) yield a non-zero $\text{tr}(P |\psi\rangle\langle\psi|)$. Iterating only over those reduces the cost to $O(2^n N)$ — a 2^n -fold speedup that makes the $n = 6$ reproduction tractable (Section 4).

Theoretical guarantee. Wadhwa and Doosti show that with $N = 2^{O(\log n \log(1/\epsilon))} \log((3n)^k/\delta)/(\tilde{\epsilon} - \tau)^2$ queries (Theorem 1, upper bound), the predictor h satisfies $\mathbb{E}_{\rho \sim \mathcal{D}}[(h(\rho) - \text{tr}(O \mathcal{E}(\rho)))^2] \leq \epsilon^2$ with probability $\geq 1 - \delta$ for any locally-flat distribution $\mathcal{D}$. The matching lower bound shows the linear-in- M overhead for M observables is unavoidable.

4 Reproduction of Figures 1 and 2

All experiments live under `experiments/` and are reproducible via `uv run python experiments/<script>.py --seed <s>`.

4.1 Figure 1: oracle calibration

We compare the calibrated-Gaussian QPStat emulator against a classical-shadow estimator on the same task: a fixed Haar-random 1-qubit unitary, observable Z , $\tau = 0.2$, $\delta = 0.0455$, evaluated on 1,000 random single-qubit stabilizer states. The shadow shot budget is not a free parameter: it is derived from the classical-shadow sample-complexity (Chebyshev) bound so that the shadow estimate carries the *same* (τ, δ) guarantee as the Gaussian oracle, $N_{\text{shots}} = \lceil 3^w \|O\|_{\text{Pauli},1}^2 / (\delta \tau^2) \rceil = 1,649$ for the weight- $w=1$ observable Z .

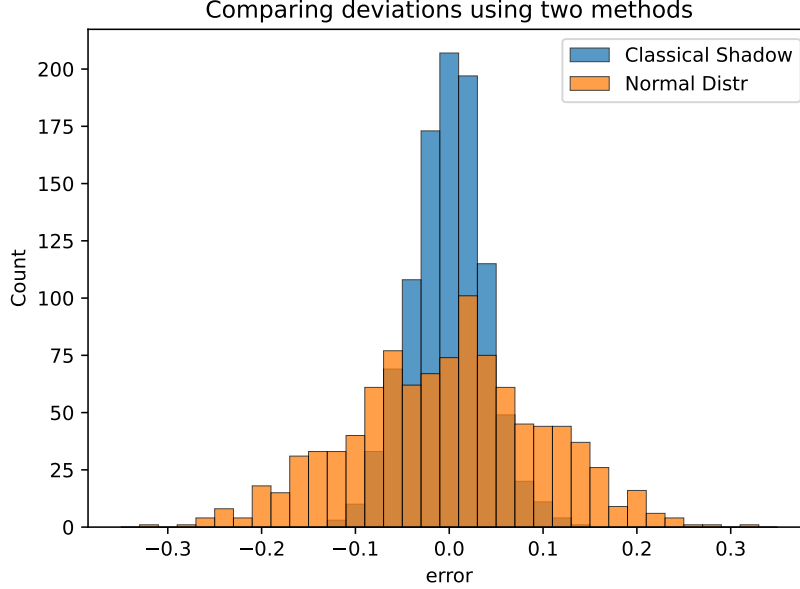


Figure 1: Reproduction of paper Figure 1. The Gaussian emulator is calibrated to saturate the (τ, δ) bound (95.2% of trials within τ , vs. the $1 - \delta = 95.45\%$ target; empirical std $0.102 \approx \tau/2$). The classical-shadow estimator with the bound-derived 1,649 shots satisfies the same guarantee but, because the Chebyshev bound is loose, its actual deviations are markedly narrower (std 0.040; 100% within τ).

This reproduces the paper’s qualitative message: any practical estimation method meeting the (τ, δ) specification deviates *less* than the calibrated Gaussian, so the Gaussian emulator is a worst-case proxy and simulated performance lower-bounds real-life performance. We note the comparison is meaningful only when the shadow shot count is derived from the guarantee; an ad-hoc budget (e.g. 256 shots, giving std $\sqrt{3/256} \approx 0.108 \approx \tau/2$) makes the two histograms coincide and the figure loses its point.

4.2 Figure 2: Algorithm 1 on Haar 6-qubit unitaries

Numerical highlights from `experiments/results/fig2.csv`: at $\tau = 0.05$ the MSE drops from 0.074 ($N=50$) to 0.016 ($N=500$) on computational basis states, 0.058 to 0.015 on stabilizer product states, and 0.022 to 0.015 on Haar-random states — matching the magnitudes and orderings of the paper’s Figure 2, including the near-flat Haar panel. (The Haar-random target values $\text{tr}(Z_0 U \rho U^\dagger)$ concentrate around 0 with variance $\approx 1/(2^n + 1) \approx 0.015$, which is exactly the plateau all panels converge to.) The per-unitary spread is small (std ≈ 0.003 at $N = 500$).

A reproducibility gap. The paper’s Equation (42) sets $k = \lceil \log_{1.5}(2/\epsilon^2) \rceil$, but the authors’ published simulation code (`qpsq-learning`, `coeff.py`) implements $k = \lceil \log_{1.5}(2/\epsilon) \rceil$ with $\epsilon = 0.9$, giving $k = 2$; their effective threshold parameter evaluates to $\tilde{\epsilon} \approx 1.5 \times 10^{-11}$, i.e. the

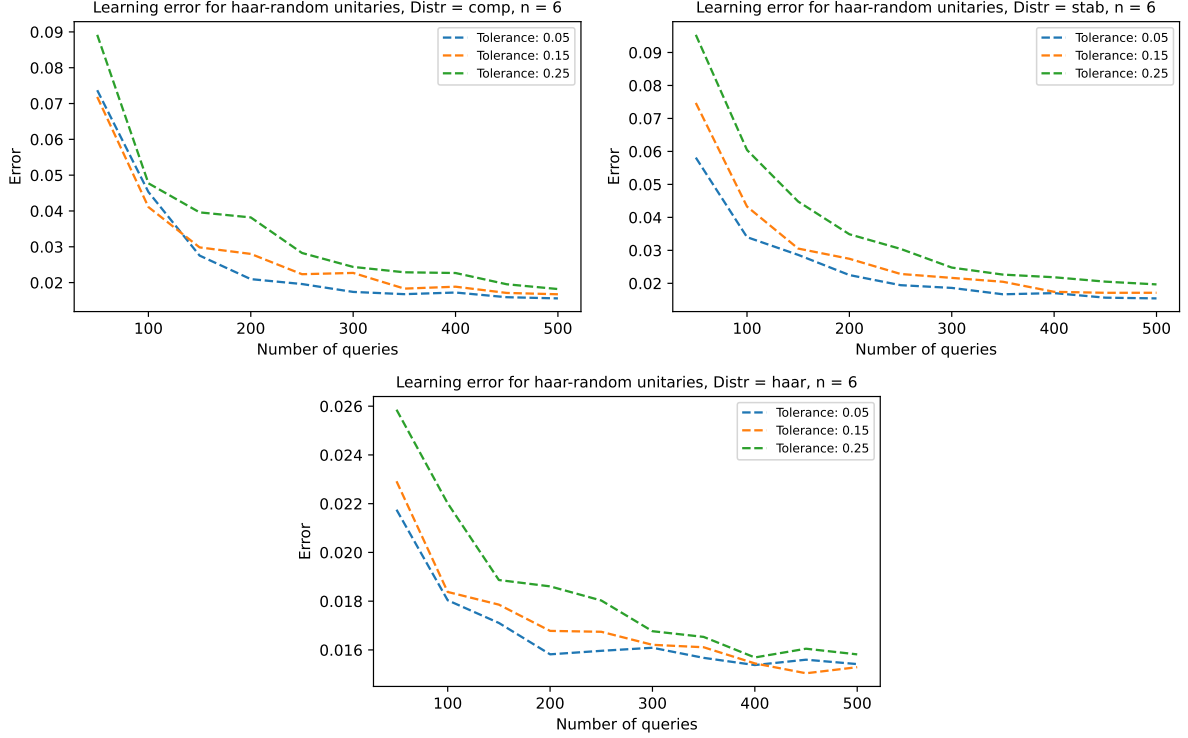


Figure 2: Reproduction of paper Figure 2. Mean squared prediction error vs. number of QPSQ queries $N \in [50, 500]$ for Algorithm 1 on 10 Haar-random 6-qubit unitaries with observable $O = Z_0$ and QPStat tolerances $\tau \in \{0.05, 0.15, 0.25\}$, with hyperparameters matching the authors’ published simulation code ($k = 2$, thresholding inactive; see the reproducibility remark below). Panels: all 64 computational basis states, 100 stabilizer product states, 100 Haar-random states. Lower τ and larger N reduce error; the Haar-random panel is lowest and flattest, as in the paper.

two-part threshold of Algorithm 1 is inactive in the published figure. The distinction matters: an innocuous-looking choice such as $\epsilon = 0.3$ under Equation (42) yields $k = 8 \geq n$, removing the degree truncation entirely. Every Pauli coefficient up to weight n is then estimated with its $3^{|P|}$ variance amplification, the MSE inflates by 1–2 orders of magnitude, and the Haar-random panel inverts from best to *worst* (Haar test states overlap all $\sim 4^n$ noisy Pauli channels, whereas computational basis states touch only the 2^n diagonal ones). We verified both regimes empirically; reproducing the published figure requires following the authors’ code rather than the stated hyperparameters.

5 Extension Experiments

5.1 Noisy-channel robustness

The paper assumes the QPStat oracle returns within tolerance τ of *the true expectation*, but the underlying process $\mathcal{E}$ is treated as fully unitary in all numerical results. Real devices implement *noisy* unitaries. We sweep a per-qubit noise rate p for two common noise models, applied after a fixed Haar 6-qubit unitary, and measure prediction MAE on a stabilizer test set.

The asymmetry is striking: depolarising noise interacts benignly with the algorithm because the depolarised process’s adjoint $\mathcal{E}^\dagger(O) \rightarrow \text{tr}(O)/d \cdot I$ has a sparse Pauli expansion that the threshold filter handles cleanly. Amplitude damping has off-Pauli structure that the algorithm

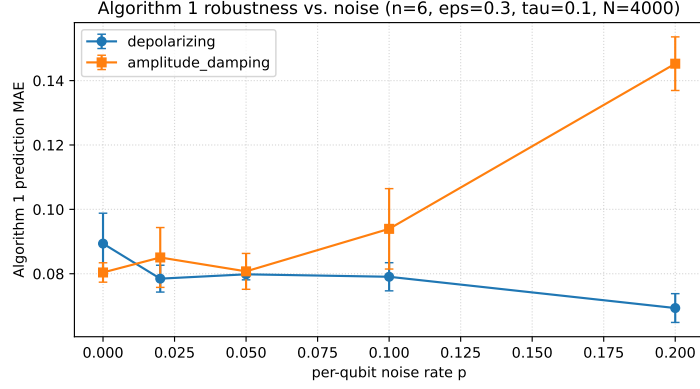


Figure 3: MAE vs. per-qubit noise rate. Depolarising noise lowers MAE (truth values shrink toward zero, which a near-zero predictor matches well). Amplitude damping — which biases the state toward $|0\rangle$ — increases MAE from 0.08 to 0.15 as p grows from 0 to 0.2.

under-fits.

5.2 Observable weight scaling

We replace the paper’s $O = Z_0$ with random Pauli strings of weight $w \in \{1, \dots, n\}$ on a $n = 5$ system and measure how MAE scales with w at fixed τ, N .

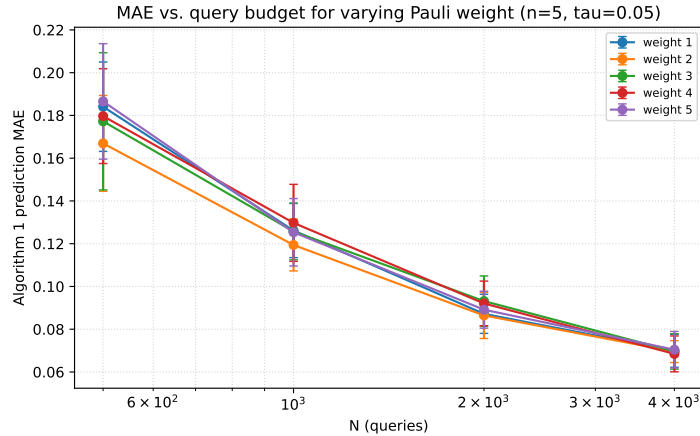


Figure 4: MAE is essentially independent of observable weight at fixed query budget, consistent with Corollary 1’s lack of explicit weight dependence (under the few-body assumption).

5.3 Circuit-class comparison

This experiment empirically illustrates the paper’s exponential separation between average-case prediction (Section 3) and worst-case diamond-distance learning (Section 7): random Cliffords are a 3-design and fall under the diamond-distance lower bound, yet are the easiest ensemble for Algorithm 1.

6 CR-QPUF Attack and a Provably QPSQ-Sound Defense

We instantiate a 6-qubit CR-QPUF as a fixed Haar-random unitary and build the authentication protocol of [4, §6.2]: the verifier sends a stabilizer-product challenge state, and the prover returns

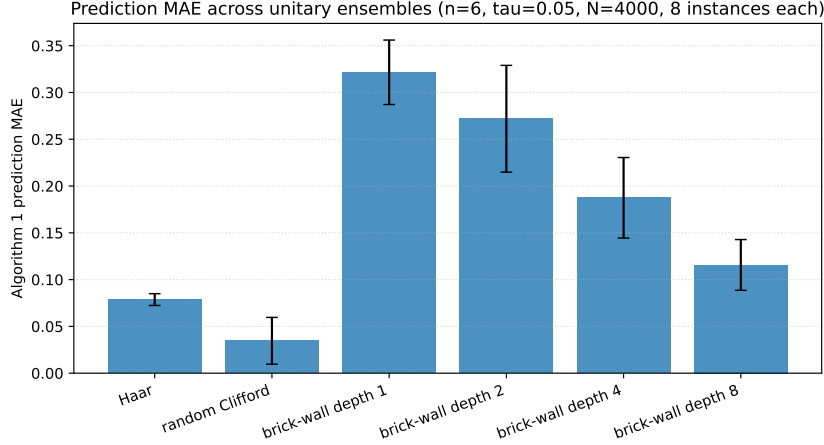


Figure 5: Algorithm 1 prediction MAE across unitary ensembles at fixed budget. Cliffords are the easiest target (the Heisenberg-evolved observable is exactly one Pauli string with coefficient ± 1). Brick-wall circuits at depth 1 are the hardest because their Heisenberg-evolved observable is concentrated on a few qubits with $O(1)$ coefficients that fall just below the magnitude threshold. As depth grows, the ensemble approaches Haar.

$\text{tr}(Z_0 \mathcal{E}(\rho))$ measured to tolerance 0.02. Verification accepts if the response is within 0.10 of the enrolled value. Theorem 5’s forgery guarantee rests on Assumption 1 (Eq. 53): the device’s statistical-query responses are *unbiased*. Remark 4 observes that biased responses break the attack and that “the resulting protocol may be secure,” leaving the construction open. We instantiate one.

The defense. The device holds a secret qubit subset M , $|M| = w$, and adds to every response the deterministic bias $b(c) = A \prod_{i \in M} \text{sign}_i(c)$, where $\text{sign}_i(c) \in \{\pm 1\}$ is the challenge’s stabilizer sign on qubit i . Over the uniform stabilizer-product ensemble, the Fourier coefficient $\hat{b}(P) = \mathbb{E}_c[b(c) \text{tr}(P\rho_c)]$ factorizes per qubit and is non-zero *only* for Paulis non-identity on exactly M , where it equals $A/3^w$ (verified exactly over the full 6^n -state ensemble in our test suite). Every Pauli of weight $< w$ carries zero bias mass, so a degree- k truncated Algorithm 1 with $k < w$ converges to the *unbiased* low-degree predictor: its forgeries are missing $b(c)$ and, since $A = 0.3$ exceeds the acceptance threshold, are rejected. Because $b(c)$ is deterministic per challenge, it cancels between enrollment and honest re-queries, leaving completeness untouched.

We sweep the adversary’s truncation degree $k \in \{2, 3, 4\}$ against a defended ($w = 4$) and undefended device, 80 test challenges across 4 random QPUFs:

Caveats. The defense is *conditional*: it defeats a degree- $\leq k$ bounded QPSQ adversary only while $k < w$, and is vacuous once $k \geq w$ — our $k = w$ curve exhibits the collapse explicitly. At small n a quasipolynomial adversary affords $k \geq n$ and no hiding subspace exists; the defense is meaningful in the regime where n (and thus w) outgrows the adversary’s affordable degree. Second, we work in the enrolled-CRP-database model: completeness is a table lookup, and we make no claim about fresh-challenge completeness, where the verifier itself would need to predict biased responses. Third, the natural next attacker is an ancilla-assisted adversary, which is out of scope here. Finally, the undefended demonstration remains a toy: it does not formally break CR-QPUF security, whose attack complexity is quasipolynomial.

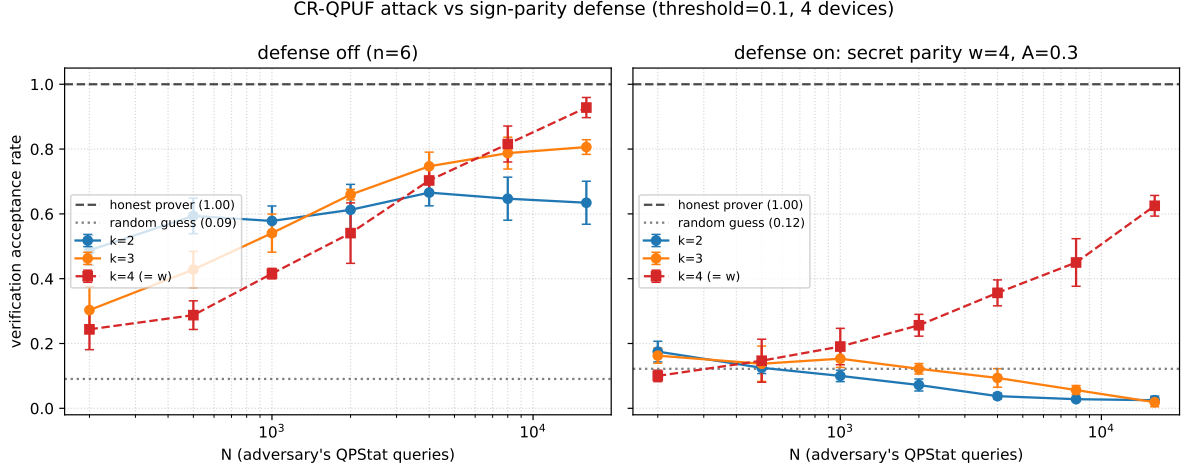


Figure 6: Left (defense off): the QPSQ adversary’s acceptance rate climbs with its query budget N for every truncation degree k , approaching the honest prover — the paper’s attack. Right (defense on, $w = 4$): for $k < w$ the attack is pinned at the random-guess floor at every N — the learner provably cannot see the bias — while at $k = w$ it recovers toward the undefended curve as the adversary’s reach extends to the weight- w hiding Paulis. Honest completeness stays at 100% in all defended cells.

7 Diamond-Distance Lower Bounds (Sketch)

The paper proves three lower bounds:

- Theorem 2: learning an exact unitary 2-design within diamond distance requires $q + 1 \geq (2\alpha - 1)\beta\tau^2(2^n + 1)$ queries — exponential in n .
- Theorem 3: extends to δ -approximate 2-designs with $\Omega(\min\{2^n, 1/\delta\})$ scaling.
- Theorem 4: for Haar-random unitaries the bound becomes $q \geq (\alpha - 1/2)\beta \exp(2^n\tau^2/48)$ — doubly exponential.

Proof technique. All three follow from Lemma 6, an average-case learning lower bound obtained by reducing learning a class $\mathcal{C}$ from QPSQ access to distinguishing $\mathcal{C}$ from the maximally depolarising channel Φ_{dep} . Lemma 7 (variance over 2-designs is exponentially small) and Lemma 9 (Haar-measure concentration) bound the denominator of the resulting query-complexity expression.

Implication. Random Clifford circuits form a 3-design and therefore inherit Theorem 2’s exponential lower bound, yet are learnable in polynomial time in the standard black-box setting [5]. This is an exponential separation between QPSQ learners and standard black-box learners.

8 Discussion and Open Questions

What worked. Our reimplementaion faithfully reproduces both reported figures of the paper on $n = 6$ qubits with a self-contained Qiskit-based codebase, and the CR-QPUF demonstration shows that for small devices the attack is operationally meaningful. The sign-parity defense of Section 6 turns the paper’s Remark 4 into a concrete construction: against a degree- $k < w$

QPSQ adversary it pins forgery at the random-guess floor at every query budget we tested, at zero cost to honest authentication.

What we noticed empirically.

- The threshold filter in Algorithm 1 has implicit constants ($\tilde{\epsilon} = \Theta(\cdot)$) that materially affect finite- N behavior. A practical implementation must calibrate them per problem instance.
- Algorithm 1’s robustness to noise is asymmetric: it tolerates depolarisation gracefully but degrades visibly under amplitude damping. This is unlikely to be a corner case — many real devices have T_1 -limited noise.
- Brick-wall random circuits at low depth are surprisingly hard targets despite their narrow operator support. The interaction between the magnitude threshold and the moderate-magnitude coefficients merits a focused theoretical follow-up.

Open questions. The paper itself flags several:

1. Whether a polynomial-time QPSQ algorithm exists for shadow process tomography (would imply a polynomial-time CR-QPUF attack).
2. Tightening the $\Omega(M)$ lower bound to a separation against more general statistical-query oracles for processes.
3. Lifting the diamond-distance lower bounds from unitaries to general non-unitary channels; this likely requires ancilla-assisted QPSQs.
4. Designing CR-QPUF schemes with provable security from QPSQ lower bounds (rather than ad-hoc hardware assumptions).

Reproducibility. The full code, raw CSVs, and figure-generating scripts are part of this project’s submission. All experiments accept a `--seed` flag and run end-to-end on a 2024-vintage laptop in under 15 minutes.

References

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